HCI Foundations & Usability

Human–Computer Interaction (HCI). The discipline concerned with the design, evaluation and implementation of interactive computing systems for human use, and the study of the major phenomena surrounding them. Poor interfaces can decrease productivity and incur additional costs.

Usability. The extent to which specified users can use a product to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.

Dimensions and course goals. **ISO dimensions:** effectiveness, efficiency and satisfaction. The six course usability goals are effectiveness, efficiency, safety, utility, learnability and memorability. Specify users, tasks, conditions and measurable criteria. Goals can conflict: confirmation may improve safety but add time.

Effectiveness. Accuracy and completeness in achieving the intended goal: correct records or a completed booking, not merely visual appeal.

Efficiency. Resources expended relative to the result achieved, such as time, effort or actions. Shortcuts and streamlined workflows reduce frequent-task effort.

Satisfaction. How acceptable or pleasing use is to the user. Responsiveness, predictable behaviour and visual appeal can enhance it; appearance alone does not establish effectiveness.

Learnability. How easily new users learn to use the system, including through exploration. Clear navigation, familiar terminology, intuitive grouping, consistent patterns and contextual guidance help.

Memorability. How easily users remember how to use the system after a period of non-use. Predictable organisation and familiar conventions/metaphors/icons support resuming tasks.

Safety. Protection from dangerous conditions and undesirable consequences of errors, including recovery. Use constraints, safe defaults, clear feedback, selective confirmation for serious consequences, and undo where feasible.

Utility and usefulness. Utility is whether the product provides the functions users need. Usefulness requires both suitable functionality and sufficient usability: an easy interface cannot compensate for a missing essential function.

User Experience (UX). A person's perceptions and responses that result from the use of a product, system or service. UX includes emotion, meaning and value as well as practical interaction.

Standards, guidelines & principles.

Standards. Established definitions, specifications or frameworks with relatively high authority and specific scope. **ISO 9241-11** provides the usability framework. **WCAG** (Web Content Accessibility Guidelines) specifies web accessibility guidance and is classified as a standard in this course.

Guidelines and principles. Guidelines recommend practice, such as iOS Human Interface Guidelines, Material Design and the Nielsen/Shneiderman rules. Principles are more general design ideas.

Patterns. A design pattern describes a recurring problem, its context and a reusable solution. An anti-pattern is recurring poor practice. A dark pattern manipulates people into actions they did not intend.

Interaction design principles.

Visibility and consistency. Visibility makes available actions and relevant information discoverable. Consistency uses predictable terms, behaviour and conventions so learning transfers.

Affordance and signifier. An affordance is an action possibility arising from the object and the user's capabilities. A signifier indicates where or how to act. The course also uses "affordance" for perceived cues suggesting use. A misleading cue can create false expectations.

Feedback, mapping and constraints. Feedback communicates the result or progress of an action. Mapping connects controls to their effects. Constraints restrict possible actions, such as disabling an invalid submission.

Visibility, consistency and familiar affordances support learnability. Feedback, mapping and navigation support control. Constraints and recovery support safety.

Nielsen's 10 Usability Heuristics. Broad usability principles for design and expert inspection; apply them to the task and context.

1. **Visibility of system status.** Keep users informed through timely feedback, such as download progress, selected states and an updated cart count.
2. **Match between system and real world.** Use users' language, familiar concepts and a natural information order.
3. **User control and freedom.** Provide exits from unwanted states, cancellation and undo/redo.
4. **Consistency and standards.** Make the same words and actions mean the same things, following platform conventions.
5. **Error prevention.** Remove error-prone conditions or check them before committing an action. Preventing invalid input precedes failure.
6. **Recognition rather than recall.** Keep options, objects and needed information visible or easily retrievable so users need not remember them between steps. Visible choices and recovery controls also support decision-making and error recovery.
7. **Flexibility and efficiency of use.** Support both novices and experts, including shortcuts and frequent-task customisation.
8. **Aesthetic and minimalist design.** Remove irrelevant information that competes with information needed for the task.
9. **Help users recognise, diagnose and recover from errors.** Explain the problem in plain language and suggest a constructive remedy.
10. **Help and documentation.** Provide concise, searchable, task-focused assistance at the point of need.

Shneiderman's Eight Golden Rules. Eight interface design rules; the table maps their overlap with Nielsen's heuristics.

Shneiderman rule	Related Nielsen heuristic
Strive for consistency	Consistency and standards.
Enable frequent-user shortcuts	Flexibility and efficiency.
Offer informative feedback	Visibility of system status.
Design dialogs to yield closure	System status: clear phase completion.
Prevent errors; handle them simply	Error prevention and error recovery.
Permit easy action reversal	User control and freedom.
Support internal locus of control	User control and freedom.
Reduce short-term memory load	Recognition rather than recall.

The course matching pairs simple error handling with error prevention. Prevention avoids failure; explanation, repair and undo address its aftermath.

Human Cognition & Perception

Cognition. The mental processes involved in acquiring and using knowledge. Perception interprets sensory input, attention selects information, memory retains and retrieves it, and learning changes knowledge or skill. Reading, listening and speaking also involve cognition.

Reflective cognition. Reasoning, decision-making and problem-solving consider alternatives and consequences. Provide accessible explanations and let users save criteria or preferences.

Attention. Selective attention focuses on relevant information among competing stimuli. Divided attention allocates limited mental resources across tasks. Multitasking can reduce performance. The Stroop effect shows interference between automatic reading and naming an ink colour.

Vision. Foveal vision supports fine detail and reading. Peripheral vision is less detailed but detects motion and brightness changes. Place messages near the current focus and make critical changes distinctive.

Gestalt principles. Perceptual rules explaining how elements form organised wholes. Distinguish the grouping mechanism in the example.

Proximity. Objects close to one another are perceived as a group. Keep within-group spacing smaller than between-group spacing, or use separators to distinguish groups.

Similarity. Objects sharing visual features such as colour, shape, size or texture are perceived as related. Shared appearance is decisive; symmetric arrangement is symmetry, shared movement is common fate.

Common Fate. Objects moving together are perceived as belonging together. Shared movement, rather than static appearance, is decisive.

Continuity. We tend to perceive smooth, continuous paths or forms rather than disconnected segments, resolving ambiguity across interruptions. Closure instead completes an incomplete contour.

Closure. We mentally complete an incomplete contour to perceive a whole object. This differs from following a continuous path.

Symmetry. Symmetric elements tend to be perceived as organised, related forms, helping interpretation.

Figure/Ground. We separate a visual field into the foreground object of attention and its background. A modal foreground can focus attention, while matching buttons inside it demonstrate similarity.

Memory and cognitive load.

Iconic memory. A brief sensory store for visual information.

Echoic memory. A brief sensory store for auditory information.

Working memory. Temporary storage and manipulation of information needed for the current task.

Working-memory components. Working memory includes a central executive for control, a visuospatial sketchpad for visual/spatial information and an articulatory loop for auditory information. Rehearsal maintains information that would otherwise decay.

Semantic memory. Long-term knowledge of facts, concepts and meanings.

Procedural memory. Long-term knowledge of how to perform skills, such as typing or driving.

Episodic / autobiographical memory. Long-term memories of personal events and experiences, such as graduation.

Permastore. Very enduring memories retained over a lifetime.

Recognition versus recall. Recognition identifies information with cues. Recall retrieves it without those cues. Visible menus, familiar labels and retained input reduce memory demands.

Miller's Law. The classic 7 ± 2 refers to meaningful chunks held in short-term memory, not a maximum number of visible menu items. Meaningful categories can reduce per-step load, but extra navigation also has a cost.

Sound, speech and haptics.

Auditory perception. Intensity relates to loudness, frequency to pitch, and timbre to sound quality.

Sound alerts. Attention-grabbing alerts reduce visual attention demands, can support multitasking and reduce information needed on screen. They can also distract, are hard to ignore, have low resolution, make absolute data hard to present and can annoy others. Suitability depends on users, task and context.

Speech and haptics. Speech conveys meaning but requires processing and intelligible output. Accent, speed and noise matter. Haptics includes tactile sensation through the skin and kinaesthesia: sensing body movement and position through muscles, joints and tendons.

Models and Norman's Gulfs.

Conceptual model. The designer's high-level description of how a system is organised and operates. The system image, including controls, wording, feedback and documentation, communicates this model to users.

Mental model. A user's understanding of how the system works, used to predict results and choose actions. Users can have different or inaccurate models. Familiar appearance can invite an incorrect model.

Gulf of Execution. The gap between the user's goals and the means of achieving them through the system. It concerns forming intentions, identifying available actions and translating goals into executable actions. Reduce it with discoverable controls, meaningful labels and clear mappings.

Gulf of Evaluation. The gap between the system's actual state and the user's ability to perceive, interpret and evaluate that state against their goals. Reduce it with visible, understandable feedback that makes progress and outcomes comparable with the user's goal.

Execution versus evaluation. "How can I do this?" indicates execution difficulty. "Did it respond, and is this what I wanted?" indicates evaluation difficulty. A system failing to execute a correct command is a functional failure; it does not alone establish that the user cannot evaluate the state.

Model mismatch. A Google-like box requiring hidden Boolean syntax invites incorrect expectations. Clarify syntax and mappings, teach the model or restore familiar cues. Think-aloud reveals expectations; inspection diagnoses issues. Neither alone repairs the interface.

Information-processing models. Human processing is modelled through perception, cognitive processing and action. The Model Human Processor predicts human performance.

External and distributed cognition. External cognition uses artefacts to reduce internal memory or calculation demands. Distributed cognition treats cognition as spread across people, tools and time, tracing how information moves between representations.

Performance laws.

Fitts' Law. Predicts the movement time needed to acquire a target: $MT = a + b \log_2(1 + D/W)$. D is movement distance and W is target width along the approach. Constants a, b depend on the device and conditions. Reduce distance or enlarge the clickable target.

Equal D/W predicts equal movement time for fixed constants. Invisible hit padding increases W. A true screen edge can prevent overshoot. Colour may help noticing, but it does not change the motor term. Total response time also includes reaction time.

Hick's Law. Choice time generally increases with the number of alternatives under comparable conditions. Fewer relevant choices can reduce decision demands. Enlarging every button mainly affects pointing, not the number of choices.

Power Law of Practice. Performance usually improves rapidly during early practice, with smaller gains as practice continues. Gains are not constant or guaranteed to stop at a fixed point. Assess experienced use as well as first use when judging long-term efficiency.

Motivation and emotion.

Maslow's hierarchy. Basic needs include physiological needs and safety. Psychological needs include belonging and esteem. Self-fulfilment concerns self-actualisation. Food illustrates physiological need, recognition esteem, and personal growth self-fulfilment.

Intrinsic and extrinsic motivation. Intrinsic motivation comes from interest or satisfaction in the activity itself, such as curiosity. Extrinsic motivation comes from an external outcome, such as public recognition or a reward.

Emotional design. Visceral design concerns immediate look, feel and sound. Behavioural design concerns use, function and usability. Reflective design concerns the product's meaning and personal value.

Affective computing. Computing that recognises, interprets, responds to or expresses emotion, including emotion recognition and expressive agents. Valence describes pleasantness, while arousal describes activation. A physiological signal alone does not uniquely identify an emotion.

Interaction Design Process

User-Centred Design (UCD). Grounds design in intended users, tasks and contexts, and refines it with user evidence. Three principles: early focus on users/tasks, empirical measurement and iterative design.

Early Focus on Users and Tasks. Study who the users are, their cognitive, behavioural and attitudinal characteristics, goals and how they perform real tasks. Understand goals and contexts before prototyping; involve intended users from early design onwards.

Empirical Measurement. Observe and measure intended users' reactions and task performance through simulations or prototypes. Use this evidence to validate assumptions and inform design decisions.

Iterative Design. When evaluation reveals problems, revise the design and test it again. Refine prototypes with real-user feedback; use qualitative and quantitative evidence, prioritising high-impact usability issues. Expert inspection complements user evidence.

User involvement. User involvement can uncover needs, manage expectations and support ownership and acceptance. Users may not know or articulate what they need. Selecting the wrong users can still produce the wrong product. Participation does not guarantee technical or business success.

Lifecycle model. A model of design activities and how they relate. The basic activities are discovering requirements, designing alternatives, prototyping and evaluating. Adapt the process to the project rather than treating it as a fixed universal sequence.

Double Diamond. A design process alternating divergent exploration with convergent focus. **Discover:** research the problem and context. **Define:** synthesise findings, map user needs and articulate a clear problem statement/design brief. **Develop:** generate, prototype and test alternative solutions. **Deliver:** refine and deliver the chosen solution.

These are typical phase responsibilities. Activities may overlap and iterate in practice. Framing clarifies purpose, context and expectations across the team.

AgileUX. Integrates UX research and design with short development iterations and frequent feedback. Iteration 0 can establish research and a product vision. Design may work one or two iterations ahead. Minimal documentation supports communication, rather than eliminating it.

PACT framework. Helps designers understand users, their activities, the contexts in which those activities occur, and the technologies involved. It supports requirements analysis and context-appropriate interaction design.

PACT–People. Investigate physical/cognitive abilities, language, culture, experience, goals, needs and motivation.

PACT–Activities. Investigate purpose, frequency, interruptions, cooperation, complexity, safety risks and information needed.

PACT–Contexts. Investigate physical conditions, social relationships and organisational rules and support.

PACT–Technologies. Consider input, output and communication, including data entry, displays, printers, bandwidth and speed. Input depends on the data/instructions; printers are output technology.

Interactions between dimensions. The dimensions interact: analyse activities in context and fit technology to people and tasks. PACT guides investigation and does not replace user research. Internet cost is an economic concern distinct from technical bandwidth.

Activity implications. Frequent tasks benefit from efficient actions. Infrequent tasks need recognisable paths. Interrupted tasks need support for resuming. Collaborative tasks need awareness and coordination. Safety-critical tasks require attention to error consequences.

Other design orientations. Activity-centred design focuses on activities. Systems design focuses on components and their relationships. Genius design relies on designer judgement. No single approach, PACT or AgileUX is mandatory for every project.

Requirements & User Research

Stakeholders. People affected by a system or able to influence its requirements. All users are stakeholders, but purchasers and indirectly affected people may also be stakeholders. Research actual end users as well as other relevant groups.

Requirement. A statement specifying what an intended product is expected to do or how it will perform. Requirements evolve and should retain supporting evidence. A user story expresses a role, desired behaviour and benefit: "As a role, I want behaviour so that benefit."

Functional requirements. What the system must do, such as calculate meal calories or record an event.

Data requirements. What data must be stored and how, including type, amount, accuracy and persistence.

Environment requirements. Physical, social, organisational or technical conditions of use, including lighting and OS compatibility.

User requirements. Relevant user characteristics, skills, roles and constraints, such as guardian permission.

Usability requirements. Required task quality, expressed as measurable effectiveness, efficiency or satisfaction criteria.

UX requirements. Intended experience or feelings, such as confidence or guilt. The category can include a negative feeling.

Classification caveat. Categories can overlap. A clinician's reporting requirements do not replace research with the patients using an app.

MoSCoW prioritisation. **Must** defines the minimum usable subset. **Should** is important but the system remains usable without it. **Could** is desirable and deferrable. **Won't this time** explicitly defers work; the course also uses "want/would have" for this lowest-priority group.

Research planning. Set goals and the analysis approach first. Identify suitable participants, establish a professional relationship and informed consent, triangulate evidence, and run a pilot. Choose methods by needed evidence, participant access, investigator skills, time, budget and equipment.

Triangulation. Use multiple methods, data sources or perspectives to investigate a question. Compare agreements and disagreements instead of assuming one method validates everything.

Pilot study. A small trial before the main study that checks questions, tasks, equipment, procedures and analysis. Revise the study if the expected evidence is not obtained.

Sampling. Probability sampling gives population members known inclusion probabilities, as in random or probability-based stratified sampling. Non-probability sampling includes convenience, snowball and volunteer recruitment, with unknown inclusion probabilities. A census covers the whole population.

Sampling versus allocation. Random sampling selects people; random allocation assigns conditions. Large, enthusiastic, technically skilled samples can remain biased. Describe demographic and experience coverage. Exact demographic matching is not always necessary.

Survey versus questionnaire. A survey is the overall methodological approach, including sampling, recruitment, questions and analysis. A questionnaire is the actual question set. Paper, email or web questionnaires can be self-administered remotely.

Surveys reach dispersed populations cheaply and support population estimates with suitable sampling. Responses may be shallow or recall-biased, with difficult immediate follow-up. Anonymity depends on collection design.

Response types. Open-ended: answers in respondents' own words, giving richer qualitative data but usually requiring more time/effort and harder statistical analysis. **Closed-ended:** predefined answers, easier to compare but may limit expression. Ordered responses include Likert ratings and ranks; unordered responses include ordinary multiple-choice or binary categories. Number codes do not create numerical distances.

Question wording. Use clear, neutral wording, group related questions and explain skip logic. Avoid leading, double-barrelled, double-negative or vague-time questions. Test on intended devices.

Standard questionnaires. SUS (System Usability Scale), CSUQ, QUIS, WAMMI and USE provide established instruments. Appropriate validated scales have established reliability/validity, are inexpensive to administer and allow scores to be compared across studies/interfaces. They measure reported experience rather than automatically diagnosing specific interface faults.

Experience and response validity. Respondents need relevant interaction experience. Merely viewing a paper design does not establish a valid experienced-usability rating. Questionnaires can yield numerical ratings and qualitative open responses.

Recording evidence. Notes are flexible but lossy and biased. Audio preserves speech, photos context and video action sequences, with privacy and analysis costs. Obtain permission and separate observation from inference.

Interview. A guided conversation for exploration, requirements or evaluation. Unstructured interviews use a flexible topic guide. Semi-structured interviews combine prepared questions with probes. Fully structured interviews ask everyone the same questions in the same order, aiding comparison but limiting exploration of new topics.

Preparation and procedure. Learn the domain and guide, including the note-taker. Introduce the study, warm up, cover main questions, cool off and close. Steer politely and attend to nonverbal cues without pre-empting answers or signalling approval. Interviews provide depth but require skill and effort and depend on recall.

Focus group. A moderated group discussion, typically with 3–10 people, to explore shared issues, norms and disagreement. Balance participation. Dominant voices, status and sensitive topics can distort responses. Consensus is not independent evidence from every person.

Observation. Study what people actually do in laboratory or natural settings. Silent observation avoids conversation. Constructive interaction observes two people working and talking together. Logs, diaries and recordings support indirect observation.

Planning and limitations. Plan timing, duration, accurate/unobtrusive recording and consent. Observation risks observer bias, reactivity (changed behaviour), unclear motives and substantial analysis effort.

Overt versus covert observation. Overt: participants know they are observed. Covert: participants do not know. Consider reactivity, consent, privacy and ethics; covert does not mean automatically permissible.

Think-Aloud. Users verbalise thoughts while performing tasks, revealing reasoning, expectations and why usability problems occur. Useful early with small samples and limited resources. **Limits:** verbalising is unnatural and can slow or alter behaviour/timing. Facilitators can bias results deliberately (task choice) or inadvertently (expressions/prompts). Users may filter speech to please the evaluator or avoid looking incompetent; verbal reports are incomplete and do not directly expose all cognitive processes.

Contextual Inquiry. Observe users doing real work while asking questions about their actions and reasoning. **Context** means visiting the work setting. **Partnership** means exploring together. **Interpretation** means checking meaning together. **Focus** guides the inquiry.

Benefits and limits. This reveals tacit needs, workflows and context-specific decisions. Full dialogue is unsuitable when interruption threatens safety, such as emergency procedures. Simple form/layout optimisation may be better served by logs or A/B testing.

Ethnography. Immersion in a group's culture and everyday practices, combining observation and interviews with ongoing interpretation. Participation can vary. Both ethnography and contextual inquiry combine observation and interview, so that phrase alone does not distinguish them.

Diary study. Participants record relevant events or make entries at intervals over time. Feedback diaries directly inform researchers. Elicitation diaries prompt later interviews. Minimise recording burden.

Documentation and auto-ethnography. Documents reveal prescribed procedures and regulations, which should be compared with actual practice. Auto-ethnography systematically examines the researcher's own lived experience. It provides a situated perspective rather than representing every user.

Data Analysis & Task Modelling

Data types and preparation.

Quantitative versus qualitative. Quantitative data express quantities, such as task time, error counts and ratings. Qualitative data capture meanings through words, images and incident descriptions. Counting incidents is quantitative; describing them is qualitative.

Mixed methods. Combine quantitative and qualitative evidence, such as Likert ratings plus interviews.

Data preparation. Clean implausible values, missing data, inconsistent formats and duplicates. Retain originals and document exclusions. Numerical coding of categories does not make them continuous measurements.

Statistical analysis.

Descriptive statistics. Mean is the sum divided by count. Median is the ordered middle, averaging the two middle values for even n. Mode is the most frequent value. Standard deviation describes spread. ANOVA is an inferential test, not a descriptive statistic.

Mean-comparison tests. For one factor, use an independent-samples t test for two separate groups and a paired-samples t test for two repeated/paired conditions. For three or more levels, use one-way ANOVA for separate groups or repeated-measures ANOVA within participants. Multiple between-participant factors use factorial ANOVA; mixed between/within factors use split-plot (mixed) ANOVA. Check scale, dependence and assumptions.

Qualitative analysis.

Critical incidents. Significant events that merit detailed investigation because of their impact. Pauses or confusion may signal an incident, but an inferred emotion is not an observed fact.

Inductive and deductive analysis. Inductive analysis develops themes from data. Deductive analysis applies predetermined categories. An affinity diagram groups individual observations or statements into emerging themes: one item per note, cluster, label and consolidate duplicates. Digital grouping is also possible.

Interaction analysis. Collaborative analysis of recorded interactions between people and artefacts. Researchers log material, assemble comparable salient events and examine patterns together.

User profiles and personas.

User profile. Describes a user group's relevant background, skills, experience and constraints to guide design and recruitment.

Persona. A fictional, research-grounded representative of a typical user type, including identity, goals, behaviours, skills and background. This shared target reduces self-referential design and shifting "users". It cannot replace real test participants.

Primary persona. The main interface design target. Aim for satisfactory use by all selected personas, accommodating secondary personas' additional needs. Prioritise explicitly rather than averaging incompatible needs or assuming the least experienced person represents everyone.

Other persona roles. Other persona roles include supplementary (needs already covered), customer (purchaser), served (non-user affected by use) and negative (outside the intended target).

Scenario. A situated narrative describing a persona's goal, setting, trigger, sequence of actions and outcome. Ground scenarios in researched tasks. A persona describes who; a scenario describes that person's activity in context.

Task analysis models.

Hierarchical Task Analysis (HTA). Decompose an overall goal into tasks and subtasks, and specify plans governing their execution. Put the root at Task 0, number siblings 1, 2, 3, then substeps 1.1, 1.2. Plans specify sequence, conditions, choice, repetition or parallel action. Numbering alone is not the plan.

HTA notation and use. A terminal-task underline marks no further decomposition in the course notation. HTA guides dialogue design, manuals and training.

GOMS. Goals state intended outcomes. **Operators** are basic actions. **Methods** are sequences for achieving goals. **Selection rules** choose between alternative methods.

Use case. A stepwise description of actor–system interaction towards a goal. An essential use case separates user intentions from system responsibilities without UI detail. A detailed case adds normal, alternative and exception paths.

Use-case diagram notation. In a use-case diagram, an actor is an external role/person, organisation or system (stick figure). A use case is an ellipse, the system boundary a rectangle, and an association a solid participation line. Associations do not show time order.

Conceptual Design & Prototyping

Conceptual design.

Object/operation analysis. Extract domain objects (nouns), attributes and operations (verbs) from scenarios. Define relationships, information flow and human/system responsibilities before widgets. Exclude menus and buffers.

Interface metaphor. Uses a familiar object or activity to help users understand an unfamiliar system, such as a desktop, shopping cart or web browsing. Assess relevance, familiarity and extensibility. Metaphors can constrain innovation or import misleading cultural assumptions.

Envisionment techniques. A sketch quickly externalises ideas. A storyboard shows a person, context and event sequence. A wireframe shows one screen's structure and layout. A wireflow links screens with actions and logic. Sketches need no artistic polish.

Prototype. A manifestation of a design that users/stakeholders can interact with to investigate an idea's possibilities and limitations. Choose fidelity and scope to test the design question within time/cost constraints; visual polish is not always the priority and a dedicated prototyping tool is not required.

Fidelity and scope. Low fidelity uses simplified representations, such as paper or cards, for quick, inexpensive changes. High fidelity resembles final content, appearance or behaviour and supports detailed interaction testing. Horizontal scope covers many functions shallowly. Vertical scope implements a narrow function deeply. These are separate dimensions.

Fidelity and measurement limits. A tool alone does not determine fidelity. High fidelity supports independent exploration but costs more, can distract towards visuals and may imply completion. Paper needs sufficient detail and durability. Manual timing is not software performance.

Prototype media. Clickthroughs link screens through hotspots. Physical prototypes use tangible materials or kits. A video prototype demonstrates a simulated scenario but does not permit free interaction. In Wizard of Oz, a hidden human generates responses users believe come from the system, allowing unfinished functions to be tested.

Evaluation & Study Design

Evaluation goals and lifecycle. Evaluate functions against tasks, effects on users, unexpected problems and alternatives. Evaluate concepts, prototypes and final products throughout development and after deployment. Formative evaluation improves designs; summative evaluation assesses outcomes against criteria.

Settings. Labs support control but may be artificial. Naturalistic studies capture context and long-term use but are harder to arrange and control. Remote describes separation, not task control.

Evaluation in real use.

Field study. Investigates natural technology use to discover requirements, opportunities or effects. In-the-wild studies let prototypes be used freely in natural settings. Logs and diaries can replace constant shadowing. Plan consent, privacy and prototype breakdowns.

Analytics, A/B and models. Analytics records behaviour at scale but does not directly explain motives. A/B testing compares design versions in comparable allocated groups using defined outcome metrics. Predictive models, such as Fitts' Law, estimate a specified performance component.

User involvement in evaluation. Inspections and models need no recruited users. The course also groups analytics under evaluation without a user session, although it still uses human data. A/B uses actual users and cannot guarantee the best design.

Heuristic Evaluation. Experts inspect an interface against recognised usability principles to identify likely problems. Choose suitable domain/accessibility rules. Brief evaluators, inspect independently in broad then detailed passes, merge findings and prioritise fixes.

Recording and prioritisation. Record location, trigger, principle, impact and remedy. Prioritise benefit against cost. Experts need user/domain knowledge but may miss issues or raise false alarms. Multiple inspectors are standard; solo review has less coverage.

Severity. 0: no problem. 1: cosmetic. 2: minor. 3: major. 4: usability catastrophe requiring repair before release. Consider frequency, impact and persistence/recovery. About five experts finding 75–80% is a guideline, not a guarantee.

Cognitive Walkthrough. Experts step through a task from a novice's perspective to predict learnability through exploration and active problem-solving, rather than memorised procedures. Specify user knowledge, context, task and correct path. At each step ask: will the user form the needed subgoal, notice the action, associate it with the goal, and interpret feedback as progress?

Walkthrough versus Think-Aloud. Walkthroughs work early without recruited users but depend on evaluator skill and effort. Think-aloud captures actual users' interpretations. Inspect first, then plan user testing.

Usability testing. Observe representative users performing representative, frequent or critical tasks to identify problems and assess task performance and satisfaction. State realistic goals without naming clicks or buttons. Coverage includes important functions. Frugality avoids redundant tasks.

Procedure and measures. Standardise starting state, instructions, equipment and assistance. Define independent/assisted/partial success, failure, errors and timing boundaries. Collect success, time, errors/help use and satisfaction separately. Success rate is successes divided by attempts; report assistance. Think-aloud may alter timing and is not unaffected summative performance.

Testing roles. The facilitator guides neutrally. The prototype executor simulates responses without coaching. Quantitative collectors time/count, qualitative collectors record observations and incidents, and actors support other roles or props.

Sample size. The famous discovery heuristic is about five users finding roughly 80% of problems, not two users or a universal minimum. The lecture also discusses 3–5, typically 5–10, or testing until no new insights. Plan for roles, complexity, availability, budget and statistical purpose.

Informed consent. Voluntary agreement to participate after receiving clear, adequate information. The right to withdraw is a related safeguard, not the definition itself.

Consent documents and session order. The Participant Information Statement (PIS) explains purpose, procedures, data use/storage/access/disposal and rights. The Participant Consent Form (PCF) records signed permission within agreed limits. Inform, answer questions and obtain participant/guardian consent before testing. Pre-questionnaires precede tasks; post-questionnaires follow. An NDA protects IP and does not replace consent.

Participant welfare. Give tasks individually, allow breaks/withdrawal, stop distress, debrief and protect records. Obtain required ethics approval before collection. Evaluate the system, not the participant.

Experiment. Tests a hypothesis about relationships between variables, aiming at new knowledge with controlled, replicable procedures and statistical analysis. Product usability testing primarily informs design and may use fewer participants and more flexible control.

Variables and conditions. The **independent variable** is the manipulated factor, such as menu type. Conditions are its levels. The **dependent variable** is the measured outcome, such as selection time. Keep control variables such as labels comparable. A control condition provides a baseline. Age/experience may be measured grouping factors rather than manipulated treatments.

Experimental design. Specify how participants experience the conditions; compare the allocation approaches below.

Design	Allocation and implications
Between-subjects	Different participants experience different conditions. Random allocation helps address individual differences.
Within-subjects	The same participants experience all conditions. Counterbalance order to reduce learning, fatigue and carryover.
Matched pairs	Match relevant traits, then allocate pair members to different conditions. Matching controls only known traits.

Counterbalancing and interpretation. AB/BA and researcher-arranged Latin-square orders distribute order effects. Correlation alone does not establish cause. Statistical significance, effect size and practical value differ.

Interface Types & Collaboration

Five interaction types. Instructing: tells the system what to do (commands or menus). **Conversing:** exchanges dialogue. **Manipulating:** acts on objects. **Exploring:** moves through physical or virtual space. **Responding:** receives system-initiated information and chooses whether to respond. Interface form and interaction type differ; one form can support several types.

Direct manipulation. Continuously represents relevant objects, supports rapid, reversible incremental actions with immediate feedback, and uses physical-style actions instead of complex syntax. Touch alone does not establish direct manipulation.

Classical interfaces: CLI and GUI.

Command-Line Interface (CLI). Accepts text commands, usually typed at a command prompt, supporting precision, scripts and batch work but requiring syntax recall. Can support visually impaired users through screen readers and speech recognition; accessibility depends on implementation.

Graphical User Interface (GUI). Classical WIMP: Windows, Icons, Menus and a Pointing device. Visible cues aid novice discovery and everyday tasks; batch work may require more clicks/steps than a command or script. Interface choice depends on requirements and environment.

Post-WIMP. Uses interaction beyond classical 2D widgets. Reality-based interaction draws on naive physics, body awareness/skills, environmental awareness/skills and social awareness/skills.

Speech interfaces. Supports hands-free interaction, commonly conversing or instructing, with noise and privacy limits. Speech input alone does not imply exploring; exploring requires movement through physical/virtual space.

Interactive surfaces. Interactable surfaces used in smart watches, smart walls and artistic installations. A surface is not necessarily a body-worn computer.

Tangible interfaces. Physical objects both control and represent digital data.

Wearable interfaces. Computers embedded in body-worn accessories, clothes or skin tattoos, usable on the move. The course describes ongoing access as "always on". Input may use touch/gaze and output visual/haptic cues; wearables are not limited to watches.

Brain-Computer Interface (BCI). Maps brain activity to device control.

VR, AR and MR. VR immerses users in a simulated sensory environment. AR overlays digital information on a real-world view. MR integrates virtual objects with the real scene in real time. Terminology can overlap.

Computer-Supported Cooperative Work (CSCW). Studies collaboration and supporting technology. Same time is synchronous and different time asynchronous. Same place is co-located and different place remote. Examples: shared whiteboard (same/same), shift board (same/different), video meeting (remote/same), email (remote/different). Classify activities, not whole products.

Presence. Telepresence is feeling present at a remote location, including through a remotely controlled robot. Social presence is awareness, feeling and response towards virtually present others. Co-presence means sharing a place: the lecture emphasises physical space, while a quiz uses shared virtual space.

Coordination and awareness. Coordination organises joint action through communication, schedules, rules and shared artefacts. Awareness is knowing who is present and what they are doing. Peripheral awareness incidentally tracks nearby activity; situational awareness includes its future implications. In turn-taking, the speaker selects someone, another self-selects or the speaker continues. Detect breakdowns before repair.

Accessibility & Situated Ability

Accessibility. Enables people with diverse abilities to use a system. Contrast is foreground/background luminance difference, distinct from hue. High contrast alone does not make colour-coded meaning accessible. Add persistent labels, icons, patterns or shapes. A red versus green outline still relies on hue.

Colour use and testing. Keep colour coding consistent and culturally appropriate. Use contrast checks and colour-vision simulation for different properties, then test tasks. Avoid fine saturated-blue details with poor contrast. The designer is not every user.

SIID. Situationally-Induced Impairments and Disabilities are limitations caused by conditions such as noise, cold, glare, stress, walking, encumbrance or divided attention. Understand the effect, sense the condition, then evaluate an adaptation. Accessible design also helps in difficult lighting and with age-related changes.